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Title: Spinal Muscular Atrophies (SMA)

Introduction

Spinal muscular atrophies (SMA) include a group of neuromuscular disorders characterized by degeneration of alpha motor neurons in the spinal cord with progressive muscle atrophy, weakness, and paralysis

The most common form of SMA is due to a defect in the survival motor neuron 1 (SMN1) gene localized to 5q11.2-q13.3.

It includes a wide range of phenotypes that are classified into clinical groups on the basis of age of onset and maximum motor function achieved

- SMA 0: results in prenatal onset of weakness and death before or shortly after birth.
- SMA1: is usually before 6 months of age. Patients with SMA1 are never able to sit, and without invasive or noninvasive respiratory support, most die by 2 years of age.
- SMA 2: presents in early childhood. These patients are never able to stand, and often live significantly longer than patients with SMA1.
- SMA 3 presents in childhood or adolescence
- SMA4 is in adulthood.

Nusinersen

- Antisense oligonucleotide
- Acts as a splicing modifier and has been approved for intrathecal treatment of SMA
- Increases the production of the functional SMN protein from SMN2
- The medication is administered intrathecally via repeated lumbar punctures
- 4 loading doses administered during the first 2 months
- Maintenance administration every 4 months thereafter, presumably for life
- Patients require 6 doses in the first year, and 3 doses annually in subsequent years

Treatment Eligibility

Spinal muscular atrophy type 1, type 2, and type 3

- a. ≤ 6 months of age at time of initiation of the treatment
- b. Diagnosed with SMA type1 as defined by bi-allelic SMN1 gene deletion or point mutation
- c. 2 copies of SMN2
- d. Not on an invasive ventilatory support

2-Patients with SMA type2:

- a. Preferably less than 5 years at the time of treatment initiation
- b. Genetically confirmed SMA with ≥ 2 copies of SMN2
- c. Not on an invasive ventilatory support

3-Patients with SMA type 3:

Patients with type 3 usually ambulatory and they start to loss ambulation by the 2nd decade

- a. At any age, Preferably less than 15 year of age at time of treatment initiation

Dosage and Administration

Spinraza is administered intrathecally

- The recommended dosage is 12 mg (5 mL) per administration
- Initiate SPINRAZA treatment with 4 loading doses: the first three loading doses should be administered at 14-day intervals; the 4 th loading dose should be administered 30 days after the 3 rd dose. A maintenance dose should be administered once every 4 months thereafter.

Important Preparation and Administration Instructions

- Allow to warm to room temperature prior to administration
- Administer within 4 hours of removal from vial
- Prior to administration, remove 5 mL of cerebrospinal fluid
- Administer as intrathecal bolus injection over 1 to 3 minutes

Laboratory Testing and Monitoring to Assess Safety

- At baseline and prior to each dose, obtain a platelet count, coagulation laboratory testing, and quantitative spot urine protein testing

Adverse Reactions

The most common adverse reactions that occurred in at least 20% of SPINRAZA-treated patients and occurred at least 5% more frequently than in control patients were:

- Lower respiratory infection and constipation in patients with infantile-onset SMA
- Pyrexia, headache, vomiting, and back pain in patients with lateronset SMA

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[Back](#)



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